



CERTIFICATE OF ANALYSIS

HESS CORPORATION
10340 68th St. NW, Tioga, North Dakota,
USA-58852

V-TIC SERVICES

VTIC LAB NUMBER	W1075	FILE NUMBER	ND231079		
LOCATION	Hawkeye Gas Facility	SOURCE OF SAMPLE	Inlet Natural Gas Line Sample Valve		
PRODUCT TYPE	Natural Gas	SAMPLE(s) SUBMITTED BY	Troy Cundiff		
GRADE/SPECIFICATION		CLIENT REFERENCE			
SEAL NO	419				
HGF NG Inlet (Spot) - W1075.001					
SAMPLED DATE & TIME: 03-Mar-2023 1030 Hrs / RECEIVED DATE & TIME: 03-Mar-2023 1900 Hrs / DATE TESTED: 22-Mar-2023					
Method Number	Method Name	Min	Max	Test Results	Unit
GPA 2286	Extended Analysis for Natural Gas & Similar Gaseous Mixtures by Temperature Program Gas Chromatography				
	Full Analysis	-	-	See Attached	-

Comment: Pressure = 20 psi
Temp = 27°F Ambient
H2S = 1.5 ppm
Cylinder# 419

AUTHORISED SIGNATORY

Curtis Kendal
Branch Manager
100000000000000452
Date: 22-Mar-2023 1549 Hrs

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V-TIC Services Inc_Williston, N.D.

C10+ Extended Natural Gas Analysis Report

Sample Information

Sample Information	
Lab	V-TIC Services Inc_Williston, N.D.
Analysis Number	W1075.001
Customer	Hess Monthly (Donny)
Sample Name	Hawkeye Gas Facility Inlet Natural Gas Line
Field	
Property	Hawkeye Gas Facility
Well ID	
Meter	
Sample Point	Sample Valve
Sample Type	Natural Gas
Cylinder Number	#419
Pressure PSIG	20 psi
Temp F	27°F Ambient
Field H2S ppm	1.5 ppm
Moisture (mmcf)	lbs/MMCF
Sample Technician	Troy Cundiff
Sample Date	3/3/23 @ 10:30am
Lab Technician	Sean Weiner
Reviewed by	Curtis Kendal
Comments	File#ND231079
Methods/Calculations/Constants	-----GPA 2286, GPA 2172, GPA 2145-----
Report Date	2023-03-22 13:38:48

Component Results

Component Name	Norm Mole%	Norm Weight%	Norm Volume%	GPM (Dry) (Gal. / 1000 cu.ft.)
Nitrogen	2.249	2.422	1.171	0.249
Methane	59.446	36.655	47.692	10.132
Carbon Dioxide	0.997	1.686	0.805	0.171
Ethane	19.135	22.115	24.217	5.145
Propane	10.028	16.996	13.074	2.778
i-Butane	1.257	2.808	1.946	0.414
n-Butane	3.955	8.835	5.900	1.254
i-Pentane	0.915	2.537	1.583	0.336
n-Pentane	1.482	4.110	2.542	0.540
n-Hexane	0.437	1.447	0.850	0.181
n-Heptane	0.086	0.331	0.188	0.040
n-Octane	0.012	0.053	0.029	0.006
n-Nonane	0.001	0.005	0.003	0.001
n-Decane	0.000	0.000	0.000	0.000
Total:	100.000	100.000	100.000	21.245

Results Summary

Result	Dry	C6+
Gross Heating Value (BTU / Ideal cu.ft.)	1487.0	4924.2
Gross Heating Value (BTU / Real cu.ft.)	1495.4	
Gross Heating Value (BTU / lbm)	21640	20918
Gross Heating Value (BTU / gal.)	70386	116777
Relative Density (G), Ideal	0.8983	3.0772
Relative Density (G), Real	0.9030	

Result	Dry	C6+
Compressibility (Z) Factor	0.9944	
Total Molecular Weight	26.018	89.124
Total Vapor Pressure (psia)	3272.40	4.32
Total Volume (cu.ft. / gal.)	47.3340	23.7149

Total Component Results

Component	Weight%	Mole%	Volume%	
Nitrogen	2.422	2.249	1.171	
Methane	36.655	59.446	47.692	
Carbon Dioxide	1.686	0.997	0.805	
Ethane	22.115	19.135	24.217	
Propane	16.996	10.028	13.074	
i-Butane	2.808	1.257	1.946	
n-Butane	8.835	3.955	5.900	
i-Pentane	2.537	0.915	1.583	
n-Pentane	4.110	1.482	2.542	
n-Hexane	1.447	0.437	0.850	
n-Heptane	0.331	0.086	0.188	
n-Octane	0.053	0.012	0.029	
n-Nonane	0.005	0.001	0.003	
n-Decane	0.000	0.000	0.000	
Total:	100.000	100.000	100.000	